

COURSE CONTEXT · SUPPLEMENTARY PROGRAM

STEAM, Grades 5–9

Read alongside `_meta-context.md` and `research/steam-progression-research.md`.**Grades** 5–9, one year plan per grade**Schedule** 2-hour session every 3 days**Year length** 24 weeks**Hours / grade** ~40 sessions, ~80 hrs

1 Course Overview

Session count assumption (state this before using the plans): “Every 3 days” is read as roughly 1.6–1.7 sessions/week on a 5-day school week — **~40 sessions/year, 2 hrs each, ~80 instructional hours per grade**. If the real rotation (e.g. a lettered-day timetable) produces a different count, unit weightings scale proportionally — the ratio between units matters more than the absolute session count. Flag it if the real number falls under 32 or over 48 sessions; the per-grade plans should be re-paced.

Audience: students and teachers. This is a supplementary program, not a mandated Alberta strand — see the Alberta angle in the research file. Materials should still be print/deliver-ready for any teacher picking up the sequence.

2 Evidence Foundation

Every structural decision in this course traces to `research/steam-progression-research.md`. Read it for the full “why.” The short version, restated as course rules:

1. **Concrete before abstract**, faded one step per grade band — not skipped, not held onto too long.
2. **All five strands run every unit, every year** — coding, robotics, spatial/construction, art/design, and the engineering design cycle are never taught as separate terms.
3. **The design cycle recurs with escalating constraints**, not as a one-off unit.
4. **At least one open-ended task per unit**, even in Grade 5 — guided-build only produces instruction-followers.
5. **Every unit ends with a share-and-reflect step** — the artifact is not the finish line, explaining it is. Capstone units (Unit 4) end with a produced video, not just a live demo.

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This is not STEM with robots bolted on. The A is a required strand with its own dedicated tasks, not a paint job applied to robotics builds — see §3 Art & Media Strand. Every unit needs at least one task that does not involve a robot at all.

3 Equipment Map

The platforms on hand map onto the concrete-to-abstract fade almost exactly as the research recommends — the backbone of the multi-year sequence. Each grade’s “primary” platform is new; prior platforms return as the *sensor/logic* layer under a more complex build, not discarded.

Platform	Coding mode	Strongest use	Primary grade
Dash (Wonder Workshop)	Blockly app, very tangible	First exposure to sequence/loop/conditional; low floor for reluctant starters	Grade 5 (retired after; concepts carry forward)
micro:bit	MakeCode blocks → Python	Physical computing, sensors (accelerometer, light, radio), wearables	Grade 5–6 intro, Gr. 7+ as sensor module
Sphero	Sphero Edu → Blockly → JavaScript	Path/maze programming, first block-to-text bridge	Grade 6 intro, Gr. 7 text transition
LEGO Spike Prime	Word Blocks → Python	Build + code integration; motors, multiple sensors	Grade 7 intro, Gr. 8–9 primary
REV kits (IQ-class)	Paired with Spike/micro:bit logic	Competition-grade construction, structural rigor	Grade 8 intro, Gr. 9 capstone
Maker/construction materials	N/A — physical only	Spatial reasoning, mechanism literacy before motorizing	Every grade, every unit
3D printers (MakerBot Sketch) + Tinkercad	Not coding — digital design/fab	Turns Build Challenge/Capstone into a real design-print-test-redesign loop	Gr. 6 intro, Gr. 7 CAD gap, Gr. 8–9 custom parts

Reading the table as a coding-abstraction ladder: Dash (fully tangible) → micro:bit blocks → Sphero blocks → Sphero JavaScript / Spike Prime blocks → Spike Prime Python. No grade should jump more than one rung in a single unit.

4 Art & Media Strand

The equipment map is a coding/robotics ladder because that’s what needs platform sequencing across five years. Art doesn’t need new hardware to fade the same way — it needs a standing requirement that keeps it from being displaced by whichever robot is newest. Two rules do that.

RULE 1 — A DEDICATED NON-ROBOT ART/DESIGN TASK, EVERY UNIT

Not aesthetics applied to a robot build (a nicer paint job on the Dash attachment doesn’t count) — a task whose deliverable is a drawing, physical model, diagram, storyboard, or other visual/design artifact that stands on its own, produced with no robot involved. This can overlap with the “at least one open-ended task per unit” rule but doesn’t have to.

- **Unit 1 (Foundations):** a sketch/diagram task — students draw and label the program flow they’re about to build, or sketch the object/character a robot behavior represents, before touching a device.
- **Unit 2 (Sensors & Response):** a visual model of the environment the robot responds to — a labeled map, diagram, or scale drawing of the test environment, made before the live test.
- **Unit 3 (Build Challenge):** the design-review artifact (sketch + material list) counts, treated as a real drawn/rendered plan, not a rough note — drafting skill is most directly on display here.
- **Unit 4 (Capstone):** the video itself is the primary art/media deliverable — storyboarding and shooting/narrating it is design work, not paperwork tacked onto the end.

RULE 2 — UNIT 4 ENDS WITH A PRODUCED VIDEO

Every grade’s Unit 4 (Capstone) ends with a produced video, not a live-only demo. Storyboard, planning, filming or screen-capturing the build/program in action, and narrating the design decisions and iteration history — replacing, or where a live audience is available, supplementing the verbal-only share-and-reflect used in Units 1–3.

This is deliberate scope: Units 1–3 stay lighter-touch so the video doesn’t become routine busywork; it’s reserved for the moment in the cycle with a real design story worth documenting. Video complexity fades the same way coding mode does — a Grade 5 video can be a simple phone/tablet recording with spoken narration; by Grade 9, students handle their own shot planning, editing, and narration with minimal scaffolding.

Digital fabrication note — this changes pacing, not just materials

A print job takes 30 minutes to several hours. No unit can plan a design-print-test loop inside a single 2-hour session: build at least one session's buffer between “design finalized” and “part in hand,” run print queues overnight or between sessions, and give students a non-printing task while a print runs. With a small number of MakerBot Sketch units and a full class needing parts, stagger submissions across the unit rather than expecting everyone's part the same day.

5 Unit Archetype

Same shape, every grade, every year. Each year is four units of ~10 sessions (~20 hrs, ~6 weeks). The unit *shape* stays constant so students recognize the pattern by Grade 7 and run it with less scaffolding — what changes year to year is the platform, the constraint complexity, and how much of the process the student initiates independently.

Unit	Focus	Constraint level	Ends with
1. Foundations	Re-establish/introduce this year's primary platform; name the CT vocabulary explicitly (sequence, loop, conditional, variable, debug)	Low — single clear goal, materials given	Working artifact + share-and-reflect
2. Sensors & Response	Programming against the physical world — line-following, obstacle response, light/sound/motion triggers	Medium — stated environment to respond to	Robot/program reacts correctly to a live class test
3. Build Challenge	Construction- and spatial-reasoning-heavy; mechanism literacy tied to the platform's actuators	Medium-high — competing constraints (speed vs. stability, weight limit)	Design review — peer critique against stated constraints
4. Capstone Design Challenge	Open-ended brief, multiple valid solutions, full design cycle with minimal prompting	High — real external constraint (budget, material, HS-modeled rubric)	Public share-out + documented iteration history, captured as a produced video

This table is the *shape*; the grade files fill in the platform, the specific brief, and the LOs.

6 Grade-by-Grade Summary

Gr.	Primary platform	Coding mode	Capstone shape
5	Dash + micro:bit (intro)	Blockly (Dash), MakeCode blocks	Guided-but-open: a Dash “helper” for a stated classroom problem
6	micro:bit + Sphero (intro)	MakeCode blocks (deeper), Sphero Edu Blockly	Sphero maze/obstacle course, student-designed, written constraint set
7	Sphero (text) + Spike Prime (intro)	Sphero JavaScript, Spike Word Blocks	Open brief: build + program Spike Prime for a task, competing constraints
8	Spike Prime (primary) + REV (intro)	Spike Prime Python (scaffolded)	Peer-reviewed design-build-test-redesign with an external constraint
9	REV (capstone) + Spike Prime Python	Python default; blocks as fallback only	Independent capstone mirroring an HS CTS/CTF outcome; documented iteration history required

7 Cross-Cutting Rules

For anyone building lessons/units from this plan.

1. **Never let a unit become “coding week” then “building week.”** Every session should touch both, even

if the ratio shifts (Unit 3 leans construction-heavy, but the platform’s code should still run something by unit’s end).

2. **Not everything needs to be done on a robot.** Every unit needs its dedicated non-robot art/design task — check for it explicitly before finalizing a unit; it’s easy to let robotics crowd it out.
3. **Name the CT vocabulary out loud at first use each year,** even if it’s a repeat. Naming beats incidental exposure at every grade, not just the youngest.
4. **At least one task per unit has more than one correct answer.** If a unit’s session plan is 100% “follow these build steps,” add or swap in an open task before finalizing.
5. **The share-and-reflect step is not optional and not “if there’s time.”** Build it into the session count for the last day of every unit. Unit 4 includes video production time.
6. **Escalate one constraint dimension at a time.** Don’t introduce a budget limit, a new platform, and a new sensor type in the same unit — pick the one escalation this unit is teaching.

8 Open Items

- Rubric design for open-ended build/capstone tasks is not yet built — flagged in the research file as a separate need. Build before the Grade 5 Unit 4 capstone is delivered.
- **The Art & Media Strand and Capstone video requirement are new** — none of the 5 grade-year files reflect them yet. Each grade file’s Units 1–4 need: (a) the non-robot art/design task named explicitly per unit, and (b) Unit 4 revised to specify the video deliverable (equipment, expected length/complexity, session fit). Follow-up pass, grade by grade.
- Video equipment/software not yet confirmed (tablets on hand for coding apps may suffice for recording; editing tool unconfirmed). Confirm before writing Grade 8–9 Unit 4, where student-led editing is assumed.
- **Confirm the CAD tool paired with MakerBot Sketch.** Plans assume Tinkercad (standard free classroom pairing) — correct if the school has a different tool set up.
- Safety/tool-use procedures for hand tools in the construction strand (Grade 5+) not yet written — needed before Unit 3 of any grade is delivered to students.
- **3D printer capacity is not yet known** (unit count, typical print time). Grade plans assume queuing is needed; confirm actual capacity so Grade 7–9 units can be paced against real parts-per-week.